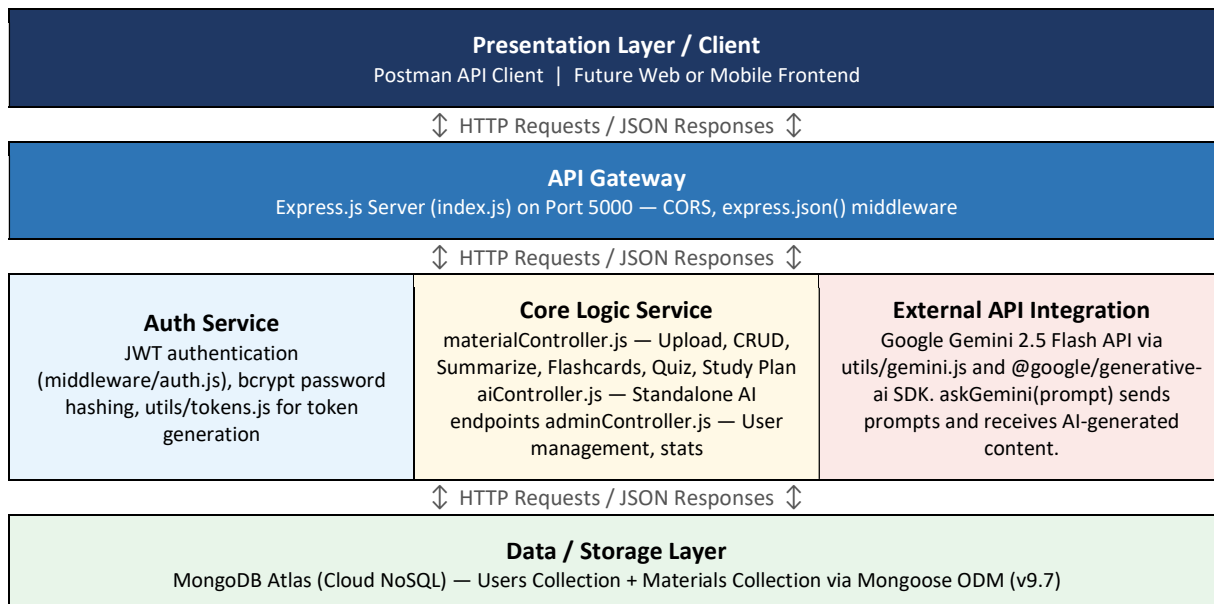


Solution Architecture

Date	29 September 2026
Team ID	6ab4ce760fc666a751b52130
Project Name	AI StudyBuddy
Maximum Marks	5 Marks

Solution Architecture Diagram

A layered architecture diagram showing how data and requests flow through the AI StudyBuddy system from client to database.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	API client for sending HTTP requests and receiving JSON responses. Currently Postman is used for testing and demonstration. Future frontend (React/Next.js) can consume the same API.	Postman, HTTP/REST
API Gateway	Express.js server (index.js) that receives all incoming requests, applies middleware (CORS, JSON parsing), and routes to appropriate controllers. Listens on Port 5000.	Express.js v4.22, cors, express.json()
Auth Service	Handles user registration, login, token refresh, and logout. Passwords hashed with bcrypt. JWT access tokens (15 min) and refresh tokens (7 days) generated via utils/tokens.js. Role-Based Access Control via middleware (protect + adminOnly).	jsonwebtoken v9.0, bcryptjs v3.0, middleware/auth.js
Core Logic Service (Controllers)	materialController: File upload via Multer, CRUD operations, AI content generation (summarize, flashcards, quiz, study-plan). aiController: Standalone AI endpoints accepting materialId in body. adminController: List users, delete users, platform statistics.	Express route handlers, Multer v2.2, Mongoose queries
External API Integration	Google Gemini 2.5 Flash API integration via utils/gemini.js. The askGemini(prompt) function sends constructed prompts to Gemini and returns AI-generated text. Used for all four AI features.	@google/generative-ai SDK v0.24, Google Gemini 2.5 Flash model
Data / Storage Layer	MongoDB Atlas cloud database with two collections: Users (name, email, hashedPassword, role) and Materials (userId, title, content, summary, flashcards[], quiz[], studyPlan). Mongoose ODM provides schemas, validation, and query interface.	MongoDB Atlas, Mongoose v9.7, User.js and Material.js schemas